

Math 122: Exam 1

Student Name: _____

Instructions:

- Show all work: Full credit requires complete, correct steps and proper math notation.
 - Points: Each question is worth 10 points.
 - Formula Sheet: Permitted (formulas only; no worked examples).
 - Calculators: Basic, non-graphing calculators allowed.
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Problem 1

Find the exact arc length of the curve

$$y = \frac{2}{3}x^{3/2} + 2026$$

over the interval from $x = 0$ to $x = 1$.

Problem 2

Find the limit:

$$\lim_{x \rightarrow 0} \frac{2^x - 1}{\sin 2x}$$

Problem 3

Find the limit:

$$\lim_{x \rightarrow +\infty} x^2 2^{-3x}$$

Problem 4

Find the limit:

$$\lim_{x \rightarrow 0^+} (1 + 3x)^{1/x}$$

Problem 5

Find $\frac{dy}{dx}$:

$$y = \sin^{-1} \left(\frac{1}{x^3} \right)$$

Problem 6

Find $\frac{dy}{dx}$:

$$y = \tan^{-1}(x) \cdot \sec^{-1}(x)$$

Problem 7

Evaluate the integral:

$$\int \frac{dx}{1 + 16x^2}$$

Problem 8

Evaluate the integral:

$$\int \frac{x^2 dx}{\sqrt{1-x^6}}$$

Problem 9

Evaluate the definite integral:

$$\int_0^1 \frac{(\sin^{-1} x)^4}{\sqrt{1-x^2}} dx$$

Problem 10

Evaluate the integral:

$$\int \frac{e^{2x}}{1 + e^{4x}} dx$$